Research Question 1

```
import pandas as pd
import numpy as np
from scipy.stats import ttest_ind, f_oneway
from statsmodels.stats.multitest import multipletests
from statsmodels.stats.power import TTestIndPower
import statsmodels.api as sm
import seaborn as sns
import matplotlib.pyplot as plt
import pymc as pm
import arviz as az
from pathlib import Path
```

EDA

```
cdi = pd.read_csv("U.S._Chronic_Disease_Indicators__CDI___2023_Release.csv", 
cvd = cdi.rename(columns={"LocationID": "statefips", "YearStart": "year"})

cvd = (
    cvd[
        (cvd["Topic"] == "Cardiovascular Disease") &
        (cvd["DataValueType"] == "Age-adjusted Rate") &
        (cvd["LocationAbbr"] != "US") &
        (cvd["StratificationCategory1"] == "Overall") &
        (cvd["year"].between(2011, 2014))
    ]
    .rename(columns={'DataValue': 'cvd_rate'})
    [['year', 'statefips', 'Question', 'cvd_rate']]
    .dropna()
)
cvd.head()
```

	year	statefips	Question	cvd_rate
276777	2011	1	Mortality from total cardiovascular diseases	298.5
292388	2013	6	Mortality from total cardiovascular diseases	206.0
297352	2014	19	Mortality from total cardiovascular diseases	209.0
299718	2014	13	Mortality from total cardiovascular diseases	241.9
303297	2011	21	Mortality from total cardiovascular diseases	272.2

```
oz = pd.read_csv("Daily_Census_Tract-Level_Ozone_Concentrations,_2011-2014_202
oz = oz.rename(columns={'avg_ds_o3_pred': 'ozone_avg'})
oz.head()
```

	year	statefips	ozone_avg
0	2011	1	41.188464
1	2011	10	40.292219
2	2011	11	39.648261
3	2011	12	38.326634
4	2011	13	42.859011

```
# Merge with ozone
df = pd.merge(cvd, oz, on=['year','statefips'], how='inner')
df['Ozone_quartile'] = pd.qcut(df['ozone_avg'], 4, labels=['Q1','Q2','Q3','Q4
df['cvd_rate'] = pd.to_numeric(df['cvd_rate'], errors='coerce')
df['ozone_avg'] = pd.to_numeric(df['ozone_avg'], errors='coerce')
df.head()
```

	year	statefips	Question	cvd_rate	ozone_avg	Ozone_quartile
0	2011	1	Mortality from total cardiovascular diseases	298.5	41.188464	Q4
1	2013	6	Mortality from total cardiovascular diseases	206.0	40.421595	Q3
2	2014	19	Mortality from total cardiovascular diseases	209.0	35.889794	Q1
3	2014	13	Mortality from total cardiovascular diseases	241.9	38.028749	Q2
4	2011	21	Mortality from total cardiovascular diseases	272.2	41.300778	Q4

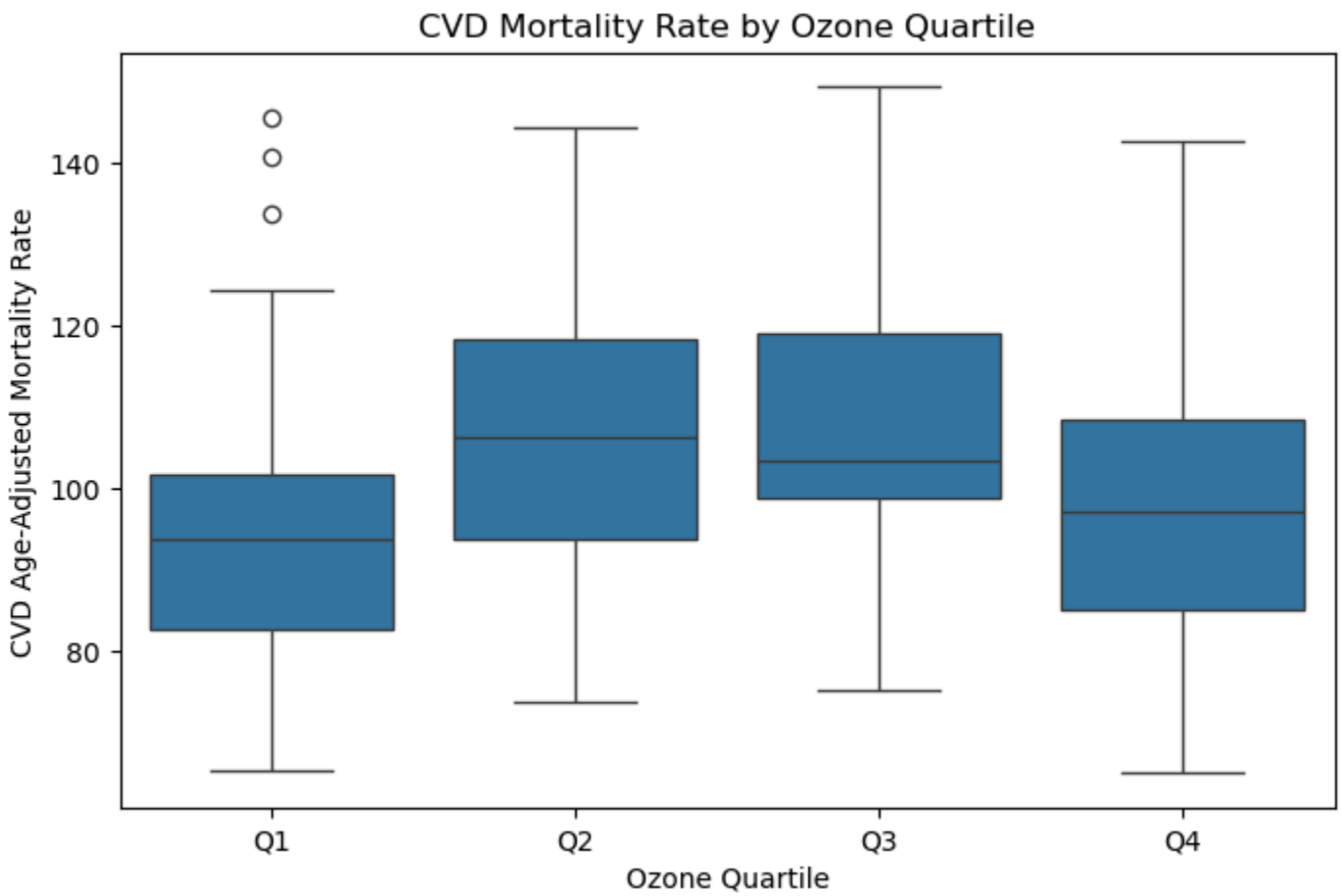
Multiple Hypothesis Testing

```
# Multiple hypothesis testing / Decision Making

# Aggregate by state-year to remove duplicates
df_agg = df.groupby(['year','statefips']).agg({
    'cvd_rate': 'mean',
    'ozone_avg': 'mean'}).reset_index()

# Assign year-specific quartiles
df_agg['Ozone_quartile'] = df_agg.groupby('year')['ozone_avg']\
                                  .transform(lambda x: pd.qcut(x, 4, labels=['

summary_stats = df_agg.groupby('Ozone_quartile', observed=True)['cvd_rate'].ag
print("=== Summary Statistics by Ozone Quartile ===")
print(summary_stats)

=== Summary Statistics by Ozone Quartile ===
                       mean        std  count
```

```
Ozone_quartile
Q1                94.775633  16.591409     52
Q2               105.693082  16.722995     48
Q3               107.594939  16.244448     48
Q4                99.797413  20.837221     48
```

Visualizations

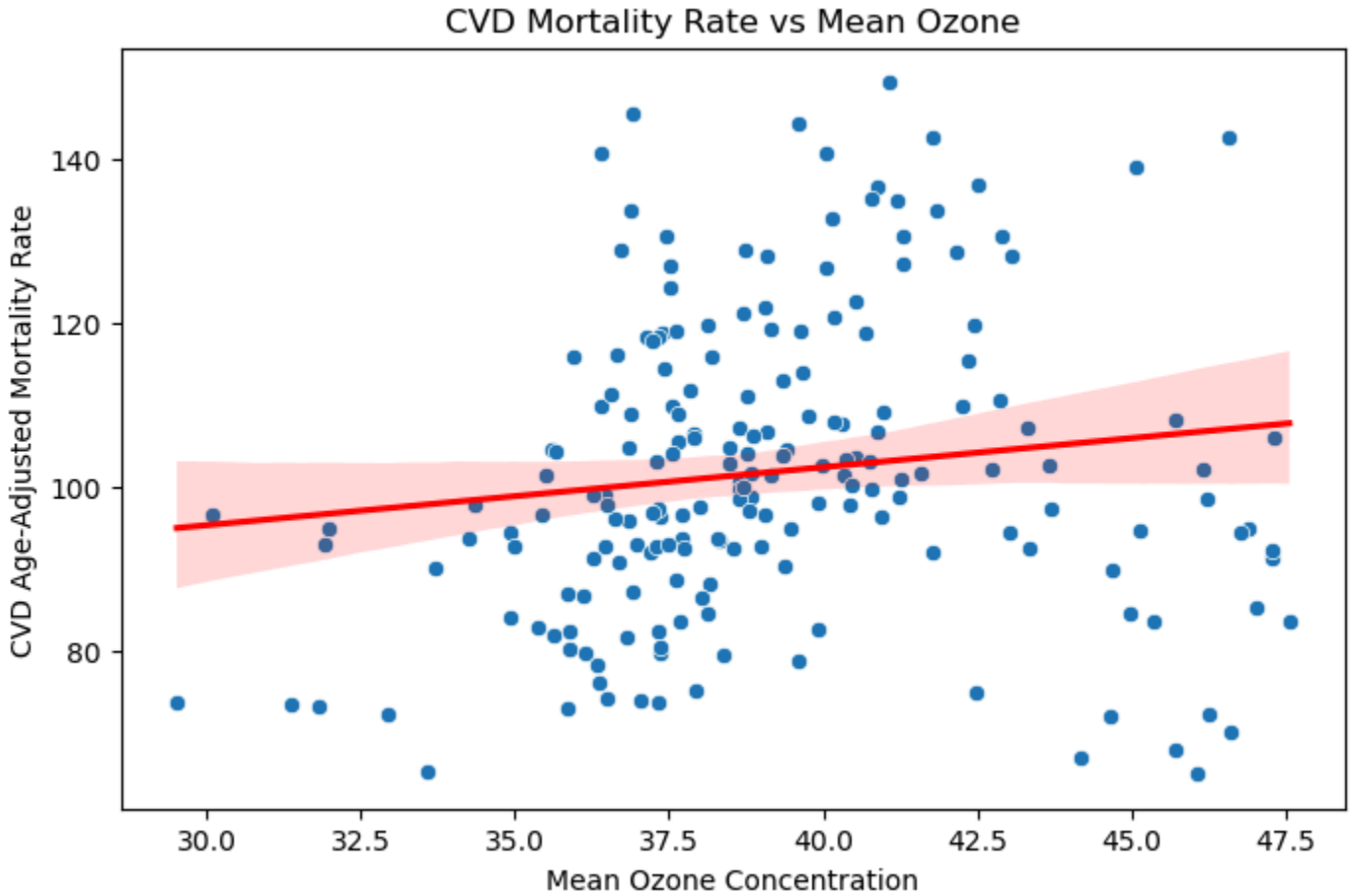
```
# Boxplot
plt.figure(figsize=(8,5))
sns.boxplot(x='Ozone_quartile', y='cvd_rate', data=df_agg)
plt.title('CVD Mortality Rate by Ozone Quartile')
plt.ylabel('CVD Age-Adjusted Mortality Rate')
plt.xlabel('Ozone Quartile')
plt.show()
```



png

```
# Scatter plot with linear fit
plt.figure(figsize=(8,5))
sns.scatterplot(x='ozone_avg', y='cvd_rate', data=df_agg)
sns.regplot(x='ozone_avg', y='cvd_rate', data=df_agg, scatter=False, color='r
plt.title('CVD Mortality Rate vs Mean Ozone')
```

```
plt.xlabel('Mean Ozone Concentration')
plt.ylabel('CVD Age-Adjusted Mortality Rate')
plt.show()
```



png

```
# Define six pairwise hypotheses
comparisons = [
    ('Q4', 'Q1'),
    ('Q3', 'Q1'),
    ('Q2', 'Q1'),
    ('Q4', 'Q2'),
    ('Q4', 'Q3'),
    ('Q3', 'Q2')
]

# Perform t-tests
results = []

for q_high, q_low in comparisons:
    group_high = df_agg[df_agg['Ozone_quartile'] == q_high]['cvd_rate']
    group_low = df_agg[df_agg['Ozone_quartile'] == q_low]['cvd_rate']
```

```
    t_stat, p_val = ttest_ind(group_high, group_low, equal_var=False)

    results.append({
        'Comparison': f"{q_high} vs {q_low}",
        't-statistic': t_stat,
        'p-value': p_val,
        'Mean_high': group_high.mean() if len(group_high) > 0 else np.nan,
        'Mean_low': group_low.mean() if len(group_low) > 0 else np.nan,
        'Std_high': group_high.std() if len(group_high) > 0 else np.nan,
        'Std_low': group_low.std() if len(group_low) > 0 else np.nan,
        'N_high': len(group_high),
        'N_low': len(group_low)
    })

results_df = pd.DataFrame(results)
results_df
```

	Comparison	t-statistic	p-value	Mean_high	Mean_low	Std_high	Std_low	N_high
0	Q4 vs Q1	1.326150	0.188154	99.797413	94.775633	20.837221	16.591409	48
1	Q3 vs Q1	3.902364	0.000175	107.594939	94.775633	16.244448	16.591409	48
2	Q2 vs Q1	3.273930	0.001470	105.693082	94.775633	16.722995	16.591409	48
3	Q4 vs Q2	-1.528800	0.129829	99.797413	105.693082	20.837221	16.722995	48
4	Q4 vs Q3	-2.044690	0.043849	99.797413	107.594939	20.837221	16.244448	48
5	Q3 vs Q2	0.565175	0.573303	107.594939	105.693082	16.244448	16.722995	48

```
# Multiple testing corrections

# Bonferroni
bonf = multipletests(results_df['p-value'], alpha=0.05, method='bonferroni')
results_df['p-value_Bonferroni'] = bonf[1]
results_df['Significant_Bonferroni'] = bonf[0]

# Benjamini-Hochberg (FDR)
fdr = multipletests(results_df['p-value'], alpha=0.05, method='fdr_bh')
results_df['p-value_FDR'] = fdr[1]
results_df['Significant_FDR'] = fdr[0]

print("\n=== Pairwise T-Test Results ===")
display_cols = ['Comparison', 't-statistic', 'p-value', 'p-value_Bonferroni',
                'Significant_Bonferroni', 'p-value_FDR', 'Significant_FDR',
                'Mean_high','Mean_low']
print(results_df[display_cols])
```

```
=== Pairwise T-Test Results ===
  Comparison  t-statistic   p-value  p-value_Bonferroni  \
0   Q4 vs Q1     1.326150  0.188154            1.000000
```

```
1   Q3 vs Q1     3.902364  0.000175            0.001051
2   Q2 vs Q1     3.273930  0.001470            0.008818
3   Q4 vs Q2    -1.528800  0.129829            0.778976
4   Q4 vs Q3    -2.044690  0.043849            0.263094
5   Q3 vs Q2     0.565175  0.573303            1.000000

   Significant_Bonferroni  p-value_FDR  Significant_FDR   Mean_high  \
0                   False     0.225785            False   99.797413
1                    True     0.001051             True  107.594939
2                    True     0.004409             True  105.693082
3                   False     0.194744            False   99.797413
4                   False     0.087698            False   99.797413
5                   False     0.573303            False  107.594939

     Mean_low
0   94.775633
1   94.775633
2   94.775633
3  105.693082
4  107.594939
5  105.693082

# Effect size & power (Q4 vs Q1)

group_Q4 = df_agg[df_agg['Ozone_quartile'] == 'Q4']['cvd_rate']
group_Q1 = df_agg[df_agg['Ozone_quartile'] == 'Q1']['cvd_rate']

mean_diff = group_Q4.mean() - group_Q1.mean()
std_Q4 = group_Q4.std()
std_Q1 = group_Q1.std()
n_Q4 = len(group_Q4)
n_Q1 = len(group_Q1)

# Pooled SD for Cohen's d
std_pooled = np.sqrt(((n_Q4-1)*std_Q4**2 + (n_Q1-1)*std_Q1**2) / (n_Q4+n_Q1-2
cohens_d = mean_diff / std_pooled

# Power analysis (two-sided)
power_analysis = TTestIndPower()
power = power_analysis.solve_power(
    effect_size=cohens_d,
    nobs1=n_Q4,
    alpha=0.05,
    ratio=n_Q1/n_Q4,
    alternative='two-sided'
)

print("\n=== Effect Size & Power for Q4 vs Q1 ===")
print(f"Mean Q4 = {group_Q4.mean():.2f}, Mean Q1 = {group_Q1.mean():.2f}")
```

```
print(f"Std Q4 = {std_Q4:.2f}, Std Q1 = {std_Q1:.2f}")
print(f"N Q4 = {n_Q4}, N Q1 = {n_Q1}")
print(f"Observed mean difference = {mean_diff:.2f}")
print(f"Cohen's d = {cohens_d:.3f}")
print(f"Power (alpha = 0.05, two-sided) = {power:.3f}")
```

```
=== Effect Size & Power for Q4 vs Q1 ===
Mean Q4 = 99.80, Mean Q1 = 94.78
Std Q4 = 20.84, Std Q1 = 16.59
N Q4 = 48, N Q1 = 52
Observed mean difference = 5.02
Cohen's d = 0.268
Power (alpha = 0.05, two-sided) = 0.263
```

Research Question 2

EDA

```
# get rid of national rows
restrict_state = ~cdi["LocationAbbr"].isin(["US", "PR"])

added_restrictions = (cdi["StratificationCategory1"].str.contains("Overall", c
| cdi["StratificationCategory1"].isna())

cdi_state = cdi[restrict_state & added_restrictions].copy()
print(cdi_state.shape)
```

```
(161345, 34)
```

```
high_bp = cdi_state[(cdi_state["Topic"] == "Cardiovascular Disease") & (cdi_st
case=False, na=False)) & (cdi_state["DataValueType"] == "Age-adjusted Prevalen

high_bp = high_bp[["YearStart", "LocationAbbr", "DataValue"]].rename(columns=
high_bp["high_bp_prev"] = pd.to_numeric(high_bp["high_bp_prev"], errors="coerc

print(high_bp.shape)
high_bp.head()
```

```
(318, 3)
```

	year	state	high_bp_prev
530586	2019	AK	32.8
531571	2015	AK	28.5
532253	2011	AK	31.0
534551	2017	AK	32.3

	year	state	high_bp_prev
534619	2013	AK	30.3

```
smoke = cdi_state[(cdi_state["Topic"] == "Tobacco") & (cdi_state["Question"].s
(cdi_state["DataValueType"] == "Age-adjusted Prevalence")].copy()

smoke = smoke[["YearStart", "LocationAbbr", "DataValue"]].rename(columns={"Yea
smoke["smoking_prev"] = pd.to_numeric(smoke["smoking_prev"], errors="coerce")


print(smoke.shape)
smoke.head()
```

```
(583, 3)
```

	year	state	smoking_prev
531128	2018	AK	18.7
531489	2012	AK	20.0
532816	2016	AK	18.7
534155	2014	AK	19.5
534181	2019	AK	17.6

```
high_bp_panel = (high_bp.groupby(["state", "year"], as_index=False)["high_bp_

smoke_panel = (smoke.groupby(["state", "year"], as_index=False)["smoking_prev

print("High BP panel:", high_bp_panel.shape)
print("Smoking panel:", smoke_panel.shape)
```

```
High BP panel: (318, 3)
Smoking panel: (583, 3)
```

```
panel = pd.merge(high_bp_panel, smoke_panel, on=["state", "year"], how="inner

panel = panel.dropna(subset=["high_bp_prev", "smoking_prev"]).copy()
print("Years:", sorted(panel["year"].unique()))
print("Num of states:", panel["state"].nunique())
panel.head()
```

```
Years: [2011, 2013, 2015, 2017, 2019, 2021]
Num of states: 52
```

	state	year	high_bp_prev	smoking_prev
0	AK	2011	31.0	22.1
1	AK	2013	30.3	22.1

	state	year	high_bp_prev	smoking_prev
2	AK	2015	28.5	18.4
3	AK	2017	32.3	20.9
4	AK	2019	32.8	17.6

```
state_codes = np.sort(panel["state"].unique())

state_to_idx = {}
for i, s in enumerate(state_codes):
    state_to_idx[s] = i

panel["state_idx"] = panel["state"].map(state_to_idx)

year_mean = panel["year"].mean()
panel["year_c"] = panel["year"] - year_mean

smoke_mean = panel["smoking_prev"].mean()
panel["smoke_centered"] = panel["smoking_prev"] - smoke_mean

scale_factor = 5.0
panel["smoke_scaled"] = panel["smoke_centered"] / scale_factor

cols_to_show = ["state", "year", "high_bp_prev", "smoking_prev", "smoke_scale
panel[cols_to_show].head()
```

	state	year	high_bp_prev	smoking_prev	smoke_scaled	year_c
0	AK	2011	31.0	22.1	0.749226	-4.974194
1	AK	2013	30.3	22.1	0.749226	-2.974194
2	AK	2015	28.5	18.4	0.009226	-0.974194
3	AK	2017	32.3	20.9	0.509226	1.025806
4	AK	2019	32.8	17.6	-0.150774	3.025806

```
state_idx = panel["state_idx"].values
smoke_scaled = panel["smoke_scaled"].values
year_c = panel["year_c"].values
y = panel["high_bp_prev"].values

n_states = panel["state_idx"].nunique()
n_states
```

52

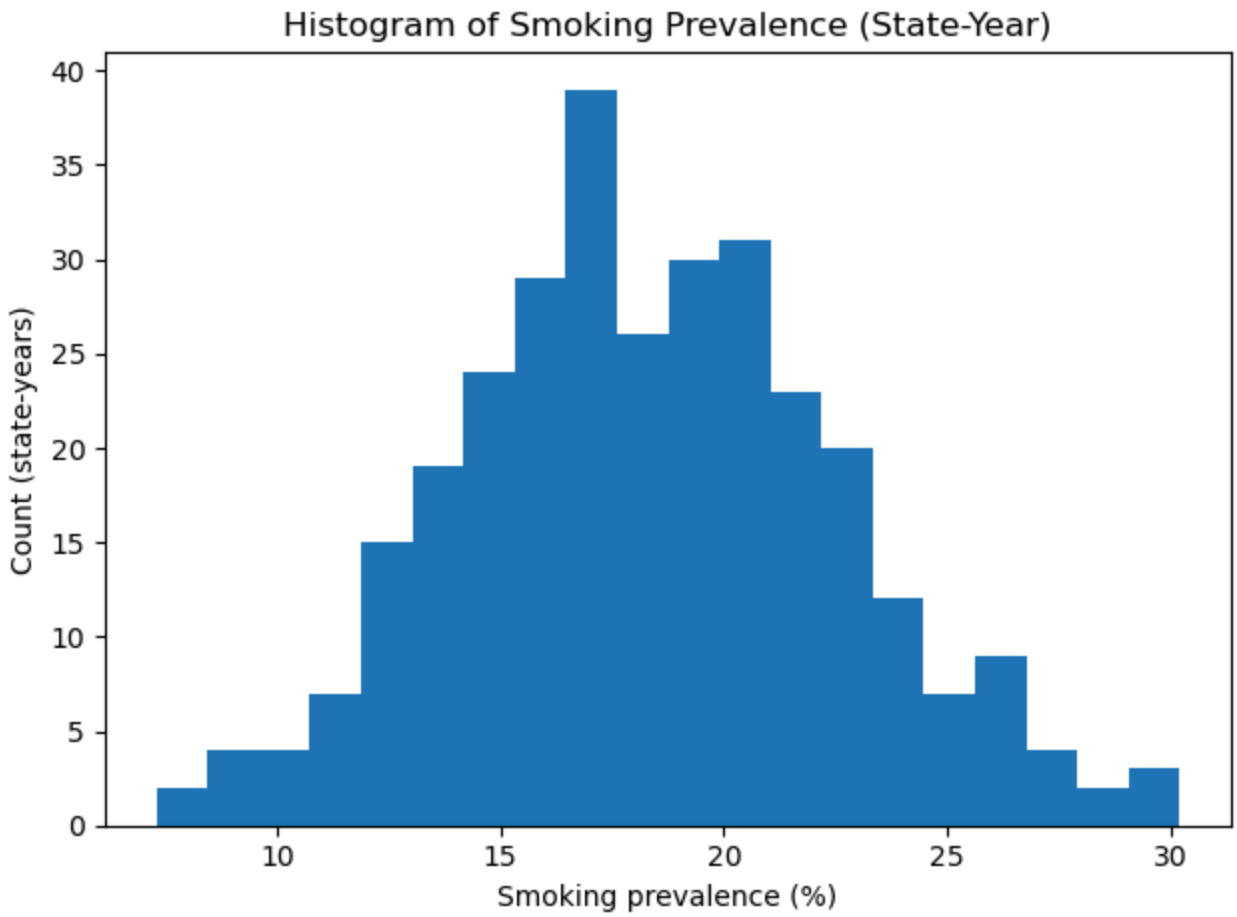
```
# Descriptive stats for key variables in the panel
desc_stats = panel[["high_bp_prev", "smoking_prev", "smoke_scaled", "year_c"]
print("Descriptive statistics for key variables:")
print(desc_stats)
```

```
# Correlation matrix between BP, smoking, and year
corr = panel[["high_bp_prev", "smoking_prev", "year"]].corr()
print("\nCorrelation matrix:")
print(corr)
```

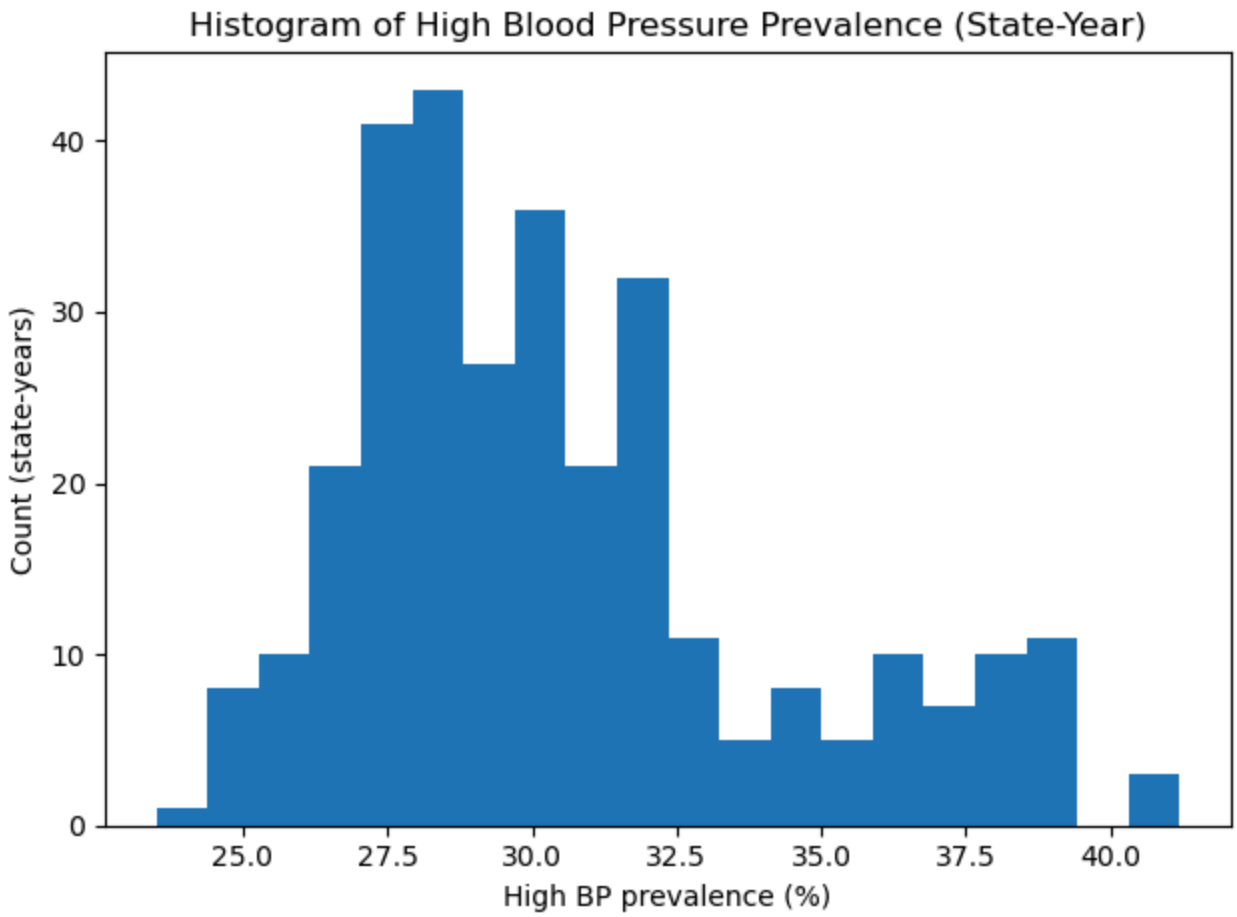
```
Descriptive statistics for key variables:
       high_bp_prev  smoking_prev  smoke_scaled        year_c
count    310.000000    310.000000  3.100000e+02  3.100000e+02
mean      30.540645     18.353871  4.268987e-16 -2.347083e-14
std        3.712782      4.245199  8.490398e-01  3.416026e+00
min       23.500000      7.300000 -2.210774e+00 -4.974194e+00
25%       27.900000     15.500000 -5.707742e-01 -2.974194e+00
50%       29.700000     18.000000 -7.077419e-02 -9.741935e-01
75%       32.075000     21.200000  5.692258e-01  3.025806e+00
max       41.200000     30.200000  2.369226e+00  5.025806e+00

Correlation matrix:
              high_bp_prev  smoking_prev      year
high_bp_prev      1.000000      0.585276  0.028355
smoking_prev      0.585276      1.000000 -0.483744
year              0.028355     -0.483744  1.000000
```

Visualizations

```
# Histogram of smoking prevalence
plt.figure()
plt.hist(panel["smoking_prev"].dropna(), bins=20)
plt.xlabel("Smoking prevalence (%)")
plt.ylabel("Count (state-years)")
plt.title("Histogram of Smoking Prevalence (State-Year)")
plt.tight_layout()
plt.show()

# Histogram of high blood pressure prevalence
plt.figure()
plt.hist(panel["high_bp_prev"].dropna(), bins=20)
plt.xlabel("High BP prevalence (%)")
plt.ylabel("Count (state-years)")
plt.title("Histogram of High Blood Pressure Prevalence (State-Year)")
plt.tight_layout()
plt.show()
```

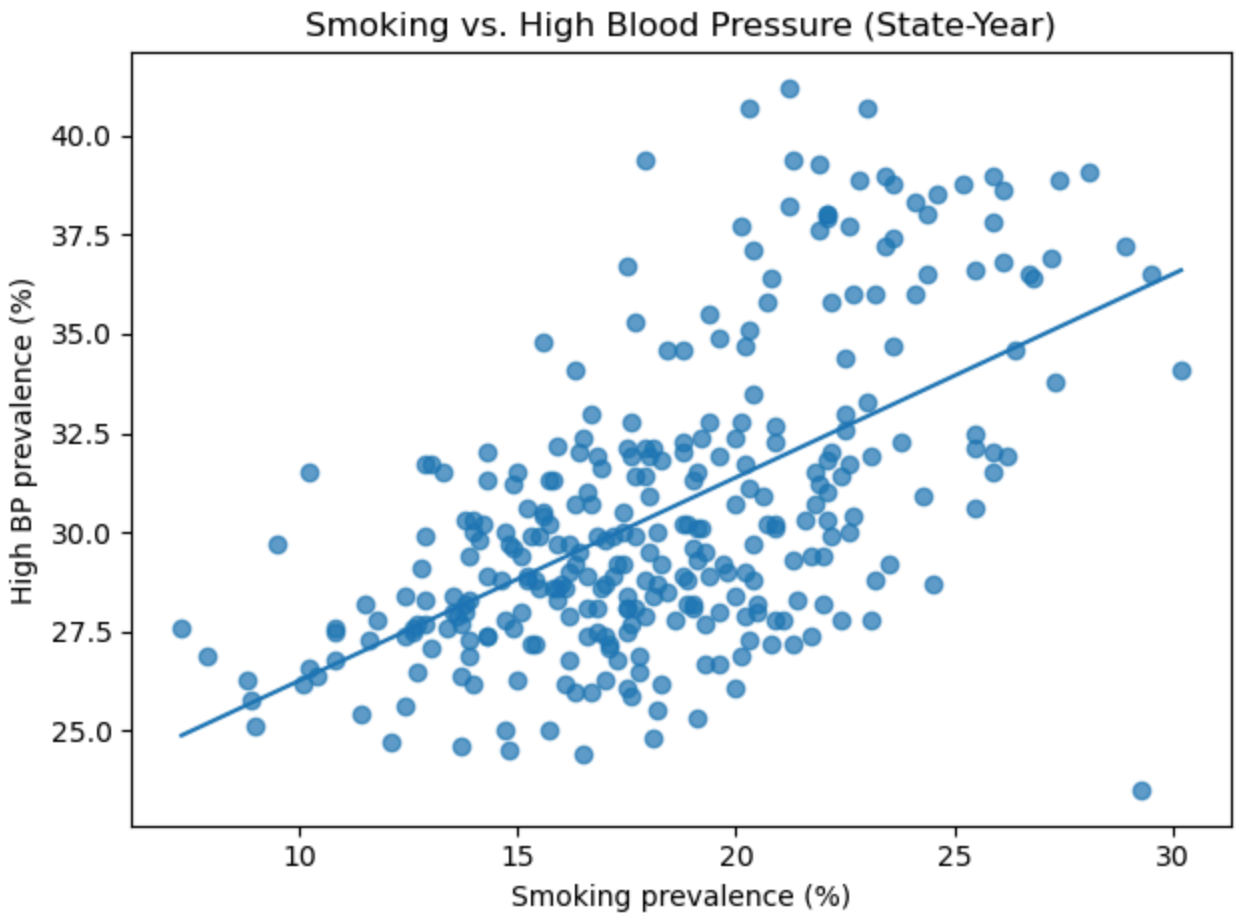

Histogram of Smoking Prevalence (State-Year)
Count (state-years)
Smoking prevalence (%)
0
5
10
15
20
25
30
35
40
10
15
20
25
30

png



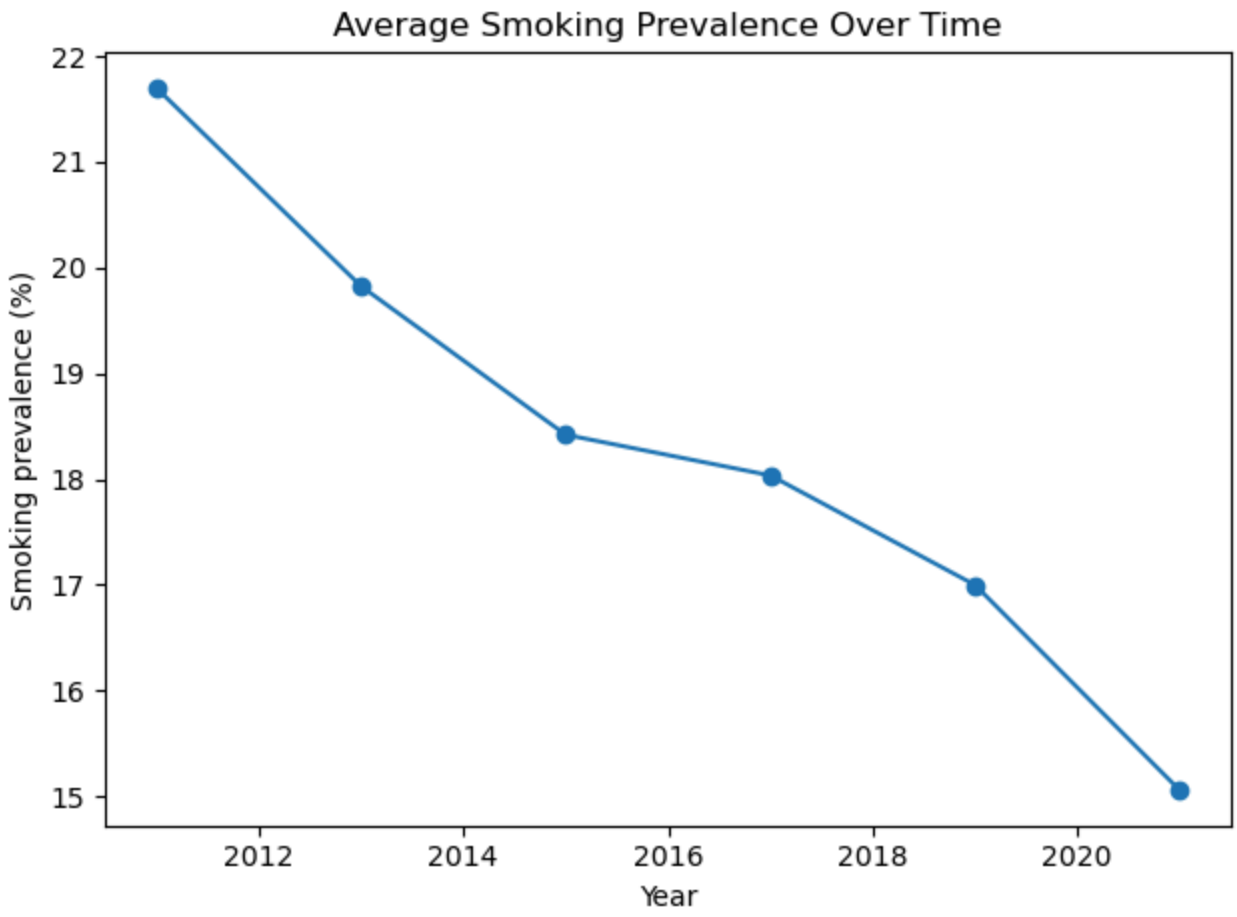
png

```
plt.figure()
plt.scatter(panel["smoking_prev"], panel["high_bp_prev"], alpha=0.7)

m, b = np.polyfit(panel["smoking_prev"], panel["high_bp_prev"], 1)
x_vals = np.linspace(panel["smoking_prev"].min(), panel["smoking_prev"].max()
plt.plot(x_vals, m * x_vals + b)

plt.xlabel("Smoking prevalence (%)")
plt.ylabel("High BP prevalence (%)")
plt.title("Smoking vs. High Blood Pressure (State-Year)")
plt.tight_layout()
plt.show()
```

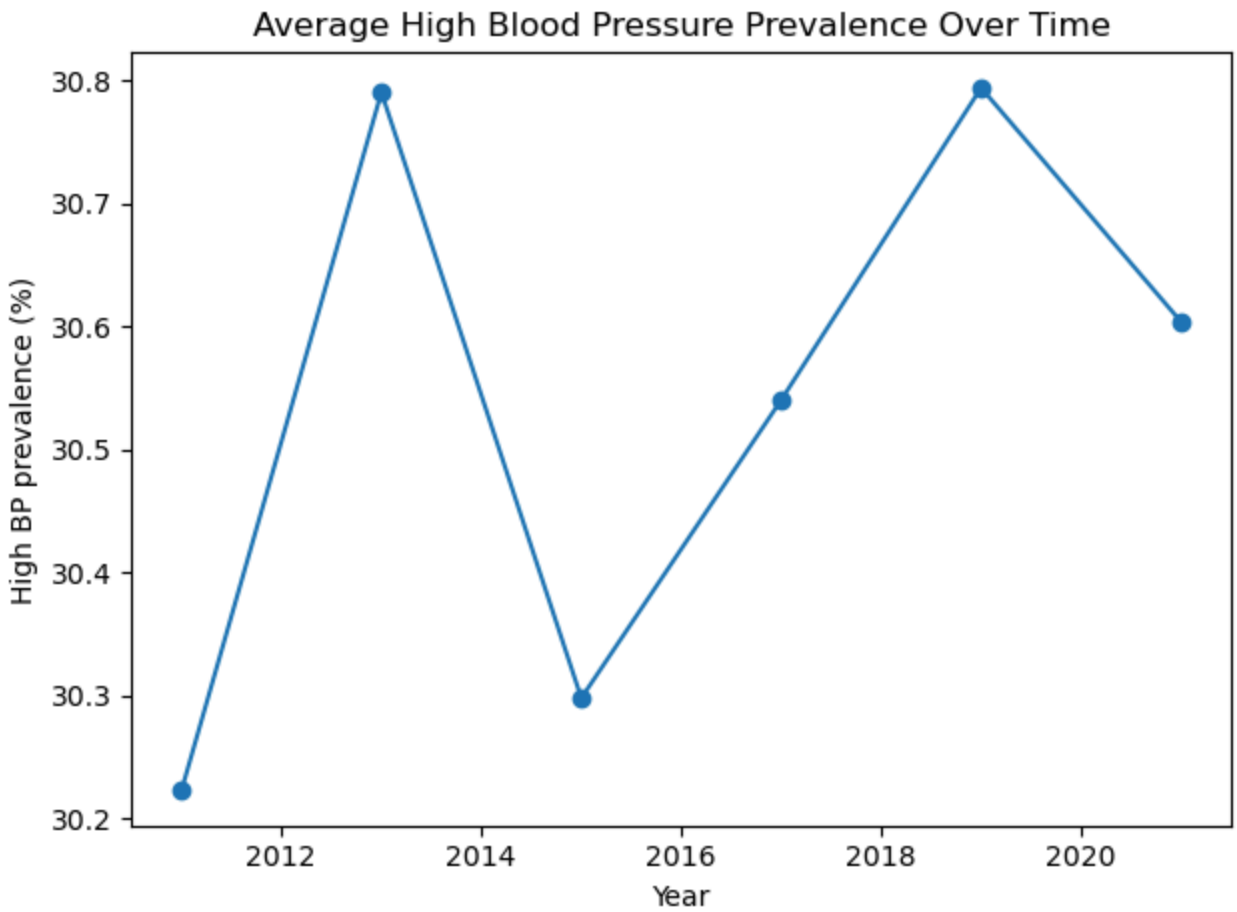


png

```
avg_by_year = panel.groupby("year")[["high_bp_prev", "smoking_prev"]].mean().

# Smoking over time
plt.figure()
plt.plot(avg_by_year["year"], avg_by_year["smoking_prev"], marker="o")
plt.xlabel("Year")
plt.ylabel("Smoking prevalence (%)")
plt.title("Average Smoking Prevalence Over Time")
plt.tight_layout()
plt.show()

# High blood pressure over time
plt.figure()
plt.plot(avg_by_year["year"], avg_by_year["high_bp_prev"], marker="o")
plt.xlabel("Year")
plt.ylabel("High BP prevalence (%)")
plt.title("Average High Blood Pressure Prevalence Over Time")
plt.tight_layout()
plt.show()
```


Average Smoking Prevalence Over Time
Smoking prevalence (%)
22
21
20
19
18
17
16
15
2012
2014
2016
2018
2020
Year

png

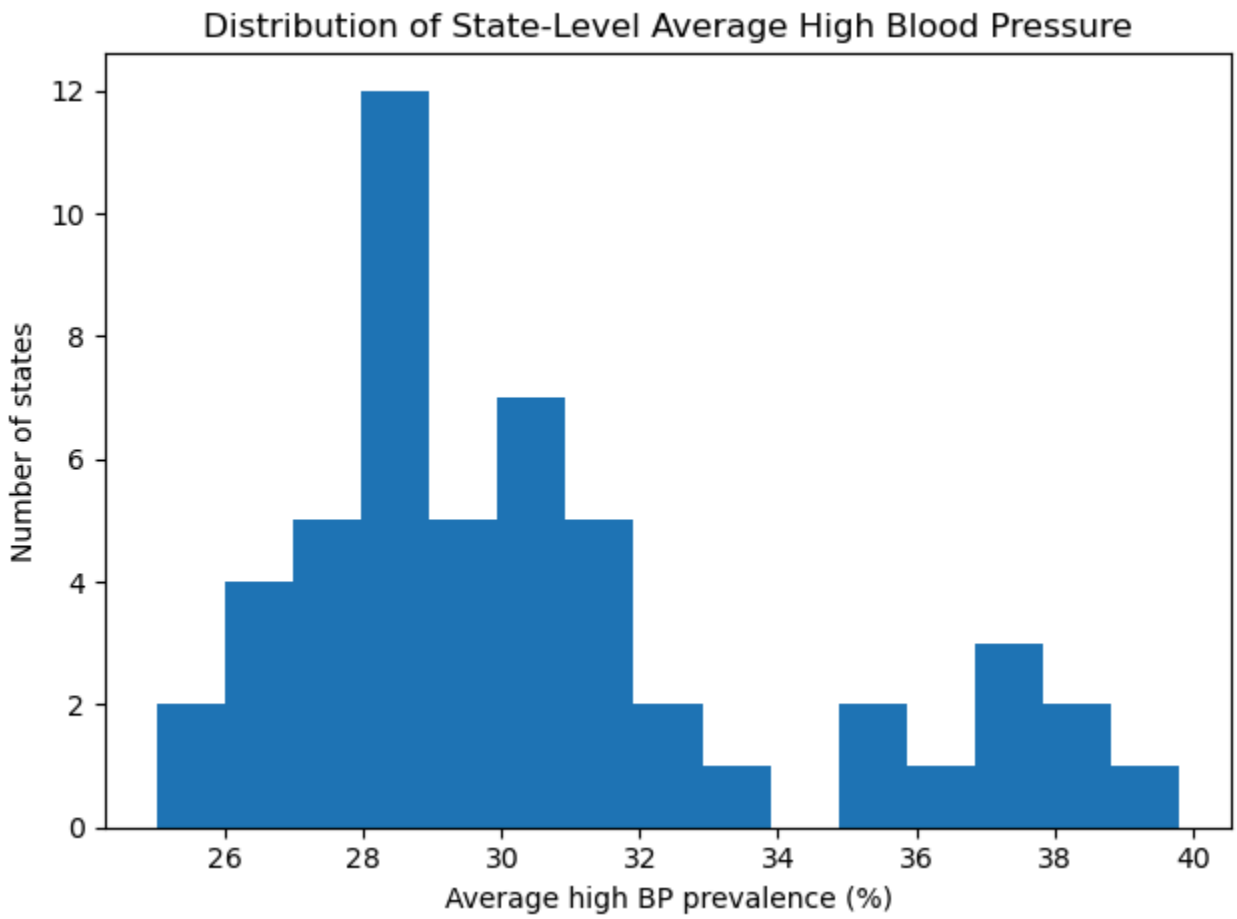


png

```
# Decompose variance of high_bp_prev
overall_var = panel["high_bp_prev"].var()

state_means = panel.groupby("state")["high_bp_prev"].mean()
between_var = state_means.var()

within_var_by_state = panel.groupby("state")["high_bp_prev"].var()
within_var = within_var_by_state.mean()

icc = between_var / (between_var + within_var)

print("Overall variance in BP prevalence:", overall_var)
print("Between-state variance:", between_var)
print("Average within-state variance:", within_var)
print("ICC (share of variance between states):", icc)

# Plot distribution of state average BP levels
plt.figure()
plt.hist(state_means.values, bins=15)
plt.xlabel("Average high BP prevalence (%)")
plt.ylabel("Number of states")
```

```
plt.title("Distribution of State-Level Average High Blood Pressure")
plt.tight_layout()
plt.show()
```

```
Overall variance in BP prevalence: 13.784750391481367
Between-state variance: 12.903036747947041
Average within-state variance: 1.2174615384615384
ICC (share of variance between states): 0.9137805540734081
```



png

Bayesian Hierarchical Model

```
with pm.Model() as bp_smoke_model:
    mu_alpha = pm.Normal("mu_alpha", mu=0.0, sigma=5.0)
    sigma_alpha = pm.HalfNormal("sigma_alpha", sigma=5.0)

    alpha_state = pm.Normal("alpha_state", mu=mu_alpha, sigma=sigma_alpha, sha

    beta_smoke = pm.Normal("beta_smoke", mu=0.0, sigma=1.0)
    gamma_year = pm.Normal("gamma_year", mu=0.0, sigma=1.0)
    sigma = pm.HalfNormal("sigma", sigma=1.0)
```

```
    mu = alpha_state[state_idx] + beta_smoke * smoke_scaled + gamma_year * yea
    y_obs = pm.Normal("y_obs", mu=mu, sigma=sigma, observed=y)

    idata = pm.sample(draws=2000, tune=2000, chains=4,target_accept=0.95, rand
```

```
Initializing NUTS using jitter+adapt_diag...
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [mu_alpha, sigma_alpha, alpha_state, beta_smoke, gamma_year, sigma]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draw
```

```
beta_samples = idata.posterior["beta_smoke"].values.flatten()

point_est = np.percentile(beta_samples, 50)
ci_low, ci_high = np.percentile(beta_samples, [2.5, 97.5])

print("Estimated change in high BP prevalence for a 5-pt increase in smoking p
print(f"Posterior median: {point_est:.2f} percentage points")
print(f"95% credible interval: [{ci_low:.2f}, {ci_high:.2f}]")
```

```
Estimated change in high BP prevalence for a 5-pt increase in smoking prevalen
Posterior median: 1.15 percentage points
95% credible interval: [0.51, 1.84]
```

```
with pm.Model() as bp_smoke_model_wide:
    mu_alpha_wide = pm.Normal("mu_alpha", mu=0.0, sigma=5.0)
    sigma_alpha_wide = pm.HalfNormal("sigma_alpha", sigma=5.0)

    alpha_state_wide = pm.Normal("alpha_state", mu=mu_alpha_wide, sigma=sigma_

    beta_smoke_wide = pm.Normal("beta_smoke", mu=0.0, sigma=5.0)
    gamma_year_wide = pm.Normal("gamma_year", mu=0.0, sigma=1.0)
    sigma_wide = pm.HalfNormal("sigma", sigma=1.0)

    mu_wide = alpha_state_wide[state_idx] + beta_smoke_wide * smoke_scaled + g

    y_obs_wide = pm.Normal("y_obs", mu=mu_wide, sigma=sigma_wide, observed=y)

    idata_wide = pm.sample(draws=2000, tune=2000, chains=4,target_accept=0.95
```

```
Initializing NUTS using jitter+adapt_diag...
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [mu_alpha, sigma_alpha, alpha_state, beta_smoke, gamma_year, sigma]
```

```
Output()

Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draw
```

```
beta_w = idata_wide.posterior["beta_smoke"].values.flatten()

point_est_w = np.percentile(beta_w, 50)
ci_low_w, ci_high_w = np.percentile(beta_w, [2.5, 97.5])

print("Wide prior results:")
print(f"Posterior median: {point_est_w:.2f} percentage points")
print(f"95% credible interval: [{ci_low_w:.2f}, {ci_high_w:.2f}]")
```

```
Wide prior results:
Posterior median: 1.34 percentage points
95% credible interval: [0.61, 2.02]
```

Posterior Summaries

```
print("Posterior summary for baseline model:")
az.summary(
    idata,
    var_names=["mu_alpha", "sigma_alpha", "beta_smoke", "gamma_year", "sigma"
    round_to=2)
```

```
Posterior summary for baseline model:
```

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
mu_alpha	30.31	0.45	29.46	31.15	0.01	0.01	7240.91	5404.16	1.00
sigma_alpha	3.08	0.35	2.46	3.76	0.01	0.00	3168.31	4666.01	1.00
beta_smoke	1.16	0.34	0.53	1.81	0.01	0.01	1015.43	2129.64	1.01
gamma_year	0.17	0.05	0.09	0.26	0.00	0.00	1172.76	2244.88	1.00
sigma	1.11	0.05	1.02	1.21	0.00	0.00	5472.60	5731.62	1.00

```
print("Posterior summary for wide beta_smoke prior model:")
az.summary(
    idata_wide,
    var_names=["mu_alpha", "sigma_alpha", "beta_smoke", "gamma_year", "sigma"
    round_to=2)
```

```
Posterior summary for wide beta_smoke prior model:
```

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
mu_alpha	30.32	0.42	29.53	31.12	0.00	0.01	10390.12	6197.54	1.00
sigma_alpha	3.01	0.35	2.36	3.64	0.01	0.00	3428.46	4604.24	1.00

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
beta_smoke	1.33	0.37	0.68	2.02	0.01	0.01	1164.01	2099.12	1.01
gamma_year	0.19	0.05	0.10	0.28	0.00	0.00	1282.52	2616.81	1.01
sigma	1.12	0.05	1.02	1.22	0.00	0.00	5573.46	5321.40	1.00